

Table 7-146. Set LDST IDT DPID Entries Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	Reserved
2h	2	LDST IDT Entry Count: Number of LDST IDT Entries being configured. Value should be >0 and not more than the lesser of the total LDST IDT table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target switch mailbox.
4h	2	Starting LDST IDT Entry Index: Index of the first LDST IDT entry being configured. An index of 0 shall designate the configuration of the 1 st entry. The starting index Plus the LDST IDT Entry Count value shall not be larger than the maximum LDST IDT entry number supported.
6h	2	Reserved
8h	Varies	LDST IDT DPID: DPID of the device for the LDST IDT entry. See Figure 7-31 and the description of interleaving in Section 7.7.2.4 . Repeats LDST IDT Entry Count number of times. - Bits[11:0]: PID of the target device - Bits[15:12]: Reserved

7.7.13.19 Get Completer ID-Based Re-Router Entries (Opcode 5712h)

This command reads the configuration of Completer ID-Based Re-Router entries.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-147. Get Completer ID-Based Re-Router Entries Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	Reserved
2h	2	Completer ID-Based Re-Router Entry Count: Number of Completer ID-Based Re-Router Entries requested. Value should be >0 and not more than the lesser of the total Completer ID-Based Re-Router table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
4h	2	Starting Completer ID-Based Re-Router Entry Index: Index of the first Completer ID-Based Re-Router entry being requested. An index of 0 shall designate the configuration of the 1 st entry. The starting index Plus the Completer ID-Based Re-Router Entry Count value shall not be larger than the maximum Completer ID-Based Re-Router entry number supported.

Table 7-148. Get Completer ID-Based Re-Router Entries Response Payload

Byte Offset	Length in Bytes	Description
0h	2	Completer ID-Based Re-Router Entry Count: Number of Completer ID-Based Re-Router Entries returned. Value should be >0 and not more than the lesser of the total Completer ID-Based Re-Router table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
2h	2	Starting Completer ID-Based Re-Router Entry Index: Index of the first Completer ID-Based Re-Router entry being returned. An index of 0 shall designate the configuration of the 1 st entry. The starting index Plus the Completer ID-Based Re-Router Entry Count value shall not be larger than the maximum Completer ID-Based Re-Router entry number supported.
4h	Varies	Completer ID-Based Re-Router Entry List[]: As defined in Table 7-149. Repeats Completer ID-Based Re-Router Entry Count number of times.

Table 7-149. Completer ID-Based Re-Router Entry

Byte Offset	Length in Bytes	Description
0h	2	Completer ID-Based Re-Router DPID[]: DPID of the requester for the Completer ID-Based Re-Router entry. - Bits[11:0]: PID of the requester - Bit[12]: Enable PCIe Segment: Indicates that the requester is in a separate PCIe segment, so the request will include the requester's and completer's segment numbers - Bits[15:13]: Reserved
2h	1	Requester PCIe Segment: PCIe Segment number for the requester. Valid only if Enable PCIe Segment is set.
3h	1	Requester Bus Number: PCIe Bus number for requester.
4h	1	- Bits[2:0]: Requester Function Number: PCIe Function number for requester - Bits[7:3]: Requester Device Number: PCIe Device number for requester

7.7.13.20 Set Completer ID-Based Re-Router Entries (Opcode 5713h)

This command sets the configuration of Completer ID-Based Re-Router entries.

This command fails with Invalid Input if access to the specified DPID is not enabled in the LAV.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- Immediate Configuration Change

Table 7-150. Set Completer ID-Based Re-Router Entries Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	Reserved
2h	2	Completer ID-Based Re-Router Entry Count: Number of Completer ID-Based Re-Router Entries being configured. Value should be >0 and not more than the lesser of the total Completer ID-Based Re-Router table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
4h	2	Starting Completer ID-Based Re-Router Entry Index: Index of the first Completer ID-Based Re-Router entry being configured. An index of 0 shall designate the configuration of the 1 st entry. The starting index given shall not be larger than the maximum Completer ID-Based Re-Router entry number supported. The starting index Plus the Completer ID-Based Re-Router Entry Count value shall not be larger than the maximum Completer ID-Based Re-Router entry number supported.
6h	Varies	Completer ID-Based Re-Router Entry List[]: As defined in Table 7-149 . Repeats Completer ID-Based Re-Router Entry Count number of times.

7.7.13.21 Get LDST Access Vector (Opcode 5714h)

This command is used by the host to query its current LAV.

This command will return Invalid Input when the requested byte range exceeds the size of the access vector buffer, as defined in [Table 7-164](#).

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-151. Get LDST Access Vector Request Payload

Byte Offset	Length in Bytes	Description
0h	4	Start Byte: Offset in bytes into Vector Data .
4h	4	Number of Bytes: Size in bytes of Vector Data requested.

Table 7-152. Get LDST Access Vector Response Payload

Byte Offset	Length in Bytes	Description
0h	4	Number of Bytes: Size in bytes of Vector Data returned.
4h	Varies	Vector Data: Excerpt of data from LDST Access Vector , defined in Table 7-153 . Excerpt begins a Start Byte and is Number of Bytes long.

Table 7-153. LDST Access Vector

Byte Offset	Length in Bytes	Description
000h	200h	LDST Access Vector: 4-Kb vector in which each bit corresponds to the associated PID (i.e., bit n represents PID n). A value of 1 in a bit position indicates that LDST and ID-Based Re-Router access to the corresponding PID is enabled. A value of 0 in a bit position indicates that access to the corresponding PID is blocked.

7.7.13.22 Get VCS LDST Access Vector (Opcode 5715h)

This command is used by the FM to query a VCS's current LAV.

This command will return Invalid Input when the requested byte range exceeds the size of the access vector buffer, as defined in [Table 7-164](#).

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-154. Get VCS LDST Access Vector Request Payload

Byte Offset	Length in Bytes	Description
0h	1	VCS ID: ID of VCS to query.
1h	3	Reserved
4h	4	Start Byte: Offset in bytes into Vector Data .
8h	4	Number of Bytes: Size in bytes of Vector Data requested.

The Get VCS LDST Access Vector Response Payload is defined in [Table 7-152](#).

7.7.13.23 Configure VCS LDST Access (Opcode 5716h)

This command is used by the FM to control access to a specified PID as reported in the LAV.

Possible Command return codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-155. Configure VCS LDST Access Request Payload

Byte Offset	Length in Bytes	Description
0h	1	VCS ID: ID of VCS to configure.
1h	1	Reserved
2h	2	• Bits[11:0]: PID: PID of LDST or Completer ID-Based Re-Router target • Bits[14:12]: Operation: Specifies which configuration to perform: — 000b = Enable PID access in the LAV — 001b = Disable PID access in the LAV — All other encodings are reserved • Bit[15]: Reserved

7.7.14 Global Memory Access Endpoint Command Set

This command set is used by a host to discover and manage the structures and devices involved in providing access to G-FAM and GIM resources.

7.7.14.1 Identify GAE (Opcode 5800h)

This command is used by the Host to query a GAE's capabilities, including maximum number of supported enabled PIDs and maximum number of simultaneous outstanding proxy operations and VendPrefixL0 support. It also reports the remaining number of proxy threads currently available.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-156. Identify GAE Request Payload

Byte Offset	Length in Bytes	Description
0h	1	Start vPPB Instance: Index of vPPB whose FAST Segment Info should be provided in the first entry in vPPB Global Memory Support Info List . The value of 0 represents the GAE. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	Number of vPPBs: Number of vPPBs in vPPB Global Memory Support Info List .

Table 7-157. Identify GAE Response Payload

Byte Offset	Length in Bytes	Description
0h	2	- Bits[11:0]: Total Number of Supported Enabled PIDs: Maximum number of PIDs that can be enabled for concurrent use with the Configure VCS PID Access command - Bit[12]: Egress Request/Ingress Completion VendPrefixL0 Supported: Indicates whether VendPrefixL0 is supported (1) or not supported (0) for Egress UIO Requests and Ingress UIO completions, as configured by the FM for this host with the Set VendPrefixL0 State command - Bit[13]: Ingress Request VendPrefixL0 Supported: Indicates whether VendPrefixL0 is supported (1) or not supported (0) for Ingress UIO requests, as configured by the FM for this host with the Set VendPrefixL0 State command - Bit[14]: G-FAM/GIM Configuration Supported: Indicates whether the switch supports (1) or does not support (0) re-configuration of the GIM Support bit with the Set FAST Segment Entry command - Bit[15]: Reserved
2h	2	Total Number of Supported Threads : Maximum number of simultaneous proxy operations supported by the GAE.
4h	2	Number of Available Threads : Remaining number of simultaneous proxy operations supported by the GAE.
6h	1	Start vPPB Instance : Index of vPPB whose FAST Segment Info is provided in the first entry in vPPB Global Memory Support Info List .
7h	1	Number of vPPBs : Number of vPPBs whose FAST Segment Info is provided in the first entry in vPPB Global Memory Support Info List .
8h	Varies	vPPB Global Memory Support Info List : List of vPPB Global Memory Support Info, as defined in Table 7-158, for the vPPBs identified with Start vPPB Instance and Number of vPPBs .

Table 7-158. vPPB Global Memory Support Info

Byte Offset	Length in Bytes	Description
0h	2	Number of FAST Segments : Total number of segments in the FAST for the specified GAE/vPPB.
2h	2	Reserved

7.7.14.2 Get PID Interrupt Vector (Opcode 5801h)

This command queries a GAE's PID interrupt vector.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-159. Get PID Interrupt Vector Request Payload

Byte Offset	Length in Bytes	Description
0h	4	Start Byte: Offset in bytes into PID Interrupt Vector.
4h	4	Number of Bytes: Size in bytes of PID Interrupt Vector requested.
8h	1	- Bit[0]: Clear on Read: A value of 1 indicates that the PID Interrupt Vector should be cleared to all 0s when this command completes. A GAE must ensure that no interrupts are lost in between capturing the current PID Interrupt Vector value for the response payload and clearing the vector's contents. - Bits[7:1]: Reserved.

Table 7-160. Get PID Interrupt Vector Response Payload

Byte Offset	Length in Bytes	Description
0h	4	Number of Bytes: Size in bytes of PID Interrupt Vector returned.
4h	Varies	PID Interrupt Vector Data: Excerpt of data from PID Interrupt Vector , defined in Table 7-161. Excerpt begins a Start Byte and is Number of Bytes long.

Table 7-161. PID Interrupt Vector

Byte Offset	Length in Bytes	Description
000h	200h	PID Interrupt Vector: 4-Kb vector in which each bit corresponds to the associated PID (i.e., bit n represents PID n). A value of 1 in a bit position indicates that the GAE has received a GAM VDM from the corresponding PID since the PID Interrupt Vector was last cleared.

7.7.14.3 Get PID Access Vectors (Opcode 5802h)

This command is used by the Host to query a GAE's current GFD Mapping Vector and VendPrefixLO Target Vector.

This command will return Invalid Input when the requested byte range exceeds the size of the access vector buffer, as defined in Table 7-164.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-162. Get PID Access Vectors Request Payload

Byte Offset	Length in Bytes	Description
0h	4	Start Byte: Offset in bytes into Vector Data .
1h	4	Number of Bytes: Size in bytes of Vector Data requested.

Table 7-163. Get PID Access Vectors Response Payload

Byte Offset	Length in Bytes	Description
0h	4	Number of Bytes: Size in bytes of Vector Data returned.
4h	Varies	Vector Data: Excerpt of data from PID Access Vector , defined in Table 7-164. Excerpt begins a Start Byte and is Number of Bytes long.

Table 7-164. PID Access Vector

Byte Offset	Length in Bytes	Description
000h	200h	GFD Mapping Vector: 4-Kb vector in which each bit corresponds to the associated PID (i.e., bit n represents PID n). A value of 1 in a bit position indicates that FAST access to the corresponding PID is enabled. A value of 0 in a bit position indicates that access to the corresponding PID is blocked.
200h	200h	VendPrefixL0 Target Vector: 4-Kb vector in which each bit corresponds to the associated PID (i.e., bit n represents PID n). A value of 1 in a bit position indicates that VendPrefixL0 access to the corresponding PID is enabled. A value of 0 in a bit position indicates that access to the corresponding PID is blocked.

7.7.14.4 Get FAST/IDT Capabilities (Opcode 5803h)

This command is used by the Host to retrieve the GAE's FAST and IDT Capabilities, per Section 7.7.2.4.

The host should re-discover the FAST/IDT Capabilities of a vPPB after a Presence Detect Changed notification has been received indicating that an adapter is present if the vPPB supports Presence Detect, or when a Link Up is detected if the vPPB does not support Presence Detect.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-165. Get FAST/IDT Capabilities Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents GAE. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	3	Reserved

Table 7-166. Get FAST/IDT Capabilities Response Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	2	Number of Segments: Number of FAST segments that are supported by this FAST/IDT. The number of entries must be 0 or a power of 2.
3h	1	FAST Segment Size - Bits[2:0]: FSegSz per Table 7-81 - Bits[7:3]: Reserved The device shall return 0h if this value has not been initialized.
4h	2	Number of IDT: Number of Interleave Device Table entries supported by this FAST/IDT.
6h	1	vPPB PID List Length: Number of PIDs assigned to this vPPB, as reported in vPPB PID List . Shall be 0 for vDSPs and vUSPs.
7h	1	- Bit[0]: Egress Request/Ingress Completion VendPrefixL0 Enabled: Indicates whether VendPrefixL0 is enabled (1) or disabled (0) for Egress UIO Requests and Ingress UIO completions - Bit[1]: Ingress Request VendPrefixL0 Enabled: Indicates whether VendPrefixL0 is enabled (1) or disabled (0) for Ingress UIO requests - Bit[7:2]: Reserved
8h	2	Reserved
Ah	8	Fabric Base: Base HPA of this FAST. The device shall return 0h if this value has not been initialized.
12h	8	Fabric Limit: Upper HPA of this FAST. The device shall return 0h if this value has not been initialized.
1Ah	Varies	vPPB PID: List of PIDs assigned to this vPPB, as defined in Table 7-167 .

Table 7-167. vPPB PID List Entry Format

Byte Offset	Length in Bytes	Description
0h	2d	- Bits[11:0]: vPPB PID: PID assigned to the vPPB - Bits[15:12]: Reserved

7.7.14.5 Set FAST/IDT Configuration (Opcode 5804h)

This command is used by the Host to set the GAE's FAST and IDT Capabilities, per [Section 7.7.2.4](#). Because the FabricBase and FabricLimit values must be aligned to the programmed FAST Segment Size, all three Host-chosen values are configured in one request.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- Immediate Configuration Change

Table 7-168. Set FAST/IDT Configuration Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	FAST Segment Size • Bits[2:0]: FSegSz per Table 7-81 • Bits[7:3]: Reserved
2h	1	• Bit[0]: Enable Egress Request/Ingress Completion VendPrefixLO: Configures whether VendPrefixLO is enabled (1) or disabled (0) for Egress UIO Requests and Ingress UIO completions • Bit[1]: Enable Ingress Request VendPrefixLO: Configures whether VendPrefixLO is enabled (1) or disabled (0) for Ingress UIO requests • Bit[7:2]: Reserved
3h	1	Reserved
4h	8	FabricBase: Base HPA of this FAST. FabricBase shall be aligned to the programmed FAST Segment Size. The value 0h will disable this FAST/IDT decoder.
Ch	8	FabricLimit: Upper HPA of this FAST. Shall be greater than FabricBase. Shall be aligned to the programmed FAST Segment Size. The value 0h will disable this FAST/IDT decoder.

7.7.14.6 Get FAST Segment Entries (Opcode 5805h)

This command reads the configuration of FAST Segment entries. The Host is responsible for mapping the GFAM range of HPAs to the appropriate number of available Segment Entries. Should the Host or the GAE have limited message payload capacity, the Host shall be responsible for breaking up the configuration operation into suitably sized requests.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-169. Get FAST Segment Entries Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	Reserved
2h	2	Seg Count: Number of FAST Segment Entries requested. Value should be >0 and not more than the lesser of the total Segment table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
4h	2	Starting Segment Index: Index of the first segment being requested. An index of 0 shall designate the configuration of the 1 st Segment, corresponding to HPA = FabricBase. The starting index Plus the Seg Count value shall not be larger than the maximum segment entry number supported.

Table 7-170. Get FAST Segment Entries Response Payload

Byte Offset	Length in Bytes	Description
0h	2	Seg Count: Number of FAST Segment Entries described in the Seg Entry_List[]. Value should be >0 and not more than the lesser of the total Segment table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
2h	2	Starting Segment Index: Index of the first segment being returned. An index of 0 shall designate the configuration of the 1 st Segment, corresponding to HPA = FabricBase. The starting index Plus the Seg Count value shall not be larger than the maximum segment entry number supported.
4h	Varies	Segment List[]: List of Segment Entries as defined in Table 7-171 .

Table 7-171. FAST Segment Entry Format

Byte Offset	Length in Bytes	Description
0h	1	Valid - Bit[0]: Valid Entry: As per Figure 7-31 - Bit[1]: GIM Segment: Segment used for GIM access - Bits[7:2]: Reserved
1h	1	Intlv - Bits[3:0]: Interleave Mode: As per Table 7-82 - Bits[7:4]: Reserved
2h	1	Gran - Bits[3:0]: Interleave Granularity: As per Table 7-83 - Bits[7:4]: Reserved
3h	1	Reserved
4h	2	DPID/IX: DPID or IDT Index, depending on Intlv field value: - Bits[11:0]: - If Intlv == 0, this is the actual DPID to which the GFAM request is to be sent. - Else, this is Index of the IDT entry that contains the DPID of the first GFD in the interleave set. See Figure 7-31 and the description of interleaving in Section 7.7.2.4. - Bits[15:12]: Reserved
6h	2	Reserved

7.7.14.7 Set FAST Segment Entries (Opcode 5806h)

This command is used by the Host to set the configuration of FAST Segment entries. The Host is responsible for mapping the GFAM range of HPAs to the appropriate number of available Segment Entries, per [Section 7.7.2.4](#). Should the host or the GAE have limited message payload capacity, the Host shall be responsible for breaking up the configuration operation into suitably sized requests.

There are two types of segments: those that access G-FAM, and those that access GIM. Valid PID targets for G-FAM segments are defined in the GMV. Valid targets for GIM segments are defined in the VTV.

This command will complete with an Invalid Input status if the requester is not authorized to access the specified ID, as advertised by the GMV or VTV.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- Immediate Configuration Change

Table 7-172. Set FAST Segment Entries Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	Reserved
2h	2	Seg Count: Number of FAST Segment Entries described in the Seg Entry_List[]. Value should be >0 and not more than the lesser of the total Segment table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
4h	2	Starting Segment Index: Index of the first segment being configured. An index of 0 shall designate the configuration of the 1 st Segment, corresponding to HPA = FabricBase. The starting index Plus the Seg Count value shall not be larger than the maximum segment entry number supported.
6h	Varies	Segment List[]: List of Segment Entries as defined in Table 7-171 .

7.7.14.8 Get IDT DPID Entries (Opcode 5807h)

This command reads the configuration of IDT entries. The Host is responsible for mapping the capacity of specific GFDs into interleaved regions of HPA. Should the Host or the GAE have limited message payload capacity, the Host shall be responsible for breaking up the configuration operation into suitably sized requests.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error

- Retry Required

Command Effects:

- None

Table 7-173. Get IDT DPID Entries Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	Reserved
2h	2	IDT Entry Count: Number of IDT Entries requested. Value should be >0 and not more than the lesser of the total IDT table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
4h	2	Starting IDT Entry Index: Index of the first IDT entry being requested. An index of 0 shall designate the configuration of the 1 st entry. The starting index Plus the IDT Entry Count value shall not be larger than the maximum IDT entry number supported.

Table 7-174. Get IDT DPID Entries Response Payload

Byte Offset	Length in Bytes	Description
0h	2	IDT Entry Count: Number of IDT Entries returned. Value should be >0 and not more than the lesser of the total IDT table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
2h	2	Starting IDT Entry Index: Index of the first IDT entry being returned. An index of 0 shall designate the configuration of the 1 st entry. The starting index Plus the IDT Entry Count value shall not be larger than the maximum IDT entry number supported.
4h	Varies	IDT DPID[]: DPID of the GFD for the IDT entry. See Figure 7-31 and the description of interleaving in Section 7.7.2.4 . Repeats IDT Entry Count number of times. • Bits[11:0]: PID of the target GFD • Bits[15:12]: Reserved

7.7.14.9 Set IDT DPID Entries (Opcode 5808h)

This command sets the configuration of IDT entries. The Host is responsible for mapping the capacity of specific GFDs into interleaved regions of HPA. Should the Host or the GAE have limited message payload capacity, the Host shall be responsible for breaking up the configuration operation into suitably sized requests.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- Immediate Configuration Change

Table 7-175. Set IDT DPID Entries Request Payload

Byte Offset	Length in Bytes	Description
0h	1	vPPB Instance: The value of 0 represents USP. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
1h	1	Reserved
2h	2	IDT Entry Count: Number of IDT Entries being configured. Value should be >0 and not more than the lesser of the total IDT table entries available, or that number of entries that can be contained in the maximum message size handled by the host and the target GAE.
4h	2	Starting IDT Entry Index: Index of the first IDT entry being configured. An index of 0 shall designate the configuration of the 1 st entry. The starting index Plus the IDT Entry Count value shall not be larger than the maximum IDT entry number supported.
6h	2	Reserved
8h	Varies	IDT DPID: DPID of the GFD for the IDT entry. See Figure 7-31 and the description of interleaving in Section 7.7.2.4 . Repeats IDT Entry Count number of times. - Bits[11:0]: PID of the target GFD - Bits[15:12]: Reserved

7.7.14.10 Proxy GFD Management Command (Opcode 5809h)

This command is used to initiate the transfer of a management command to a GFD, as defined in [Section 3.1.11.1](#).

Only one proxy request may be outstanding per target PID regardless of the number of available proxy threads. A proxy request that targets a PID with an existing outstanding proxy request shall fail with 'Invalid Input'. The command shall fail with 'Resources Exhausted' if there are no available proxy operation threads.

The GAE increments and tracks Command Sequence Number on a per-Target PID basis.

This command will complete with an Invalid Input status if the requester is not authorized to access the specified ID, as advertised by the GMV.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required
- Resources Exhausted

Command Effects:

- None

Table 7-176. Proxy GFD Management Command Request Payload

Byte Offset	Length in Bytes	Description
00h	2	- Bits[11:0]: PBR-ID: Target PID for the management command - Bits[15:12]: Reserved
02h	8	Request Address : Pointer to request message in Host memory that is formatted in the CCI Message Format as defined in Figure 7-19.
0Ah	2	Request Size : Size of the request at Request Address in bytes.
0Ch	8	Response Address : Pointer in Host memory at which the response should be written. The response shall be formatted in the CCI Message Format as defined in Figure 7-19.
14h	2	Maximum Response Size : Size of the response at Request Address in bytes.

Table 7-177. Proxy GFD Management Command Response Payload

Byte Offset	Length in Bytes	Description
0h	1	- Bits[2:0]: Command Sequence Number: Proxy thread identifier for use with Get Proxy Thread Status request - Bits[7:3]: Reserved
1h	2	Number of Available Threads : Remaining number of simultaneous proxy operations supported by the GAE.

7.7.14.11 Get Proxy Thread Status (Opcode 580Ah)

This command queries whether the GAE is tracking the specified Command Sequence Number and Target PID as 'In Progress'.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-178. Get Proxy Thread Status Request Payload

Byte Offset	Length in Bytes	Description
0h	2	- Bits[11:0]: PBR-ID: Target PID for the management command - Bits[14:12]: Command Sequence Number: Proxy thread identifier returned by Proxy GFD Management Command request - Bit[15]: Reserved

Table 7-179. Get Proxy Thread Status Response Payload

Byte Offset	Length in Bytes	Description
0h	1	- Bit[0]: In Progress: A value of 1 indicates that the GAE is tracking the specified Thread Handle. A value of 0 indicates that the GAE is not tracking the specified Thread Handle. - Bits[7:1]: Reserved.
1h	1	Reserved
2h	2	Number of Available Threads : Remaining number of simultaneous proxy operations supported by the GAE.

7.7.14.12 Cancel Proxy Thread (Opcode 580Bh)

This command effectively cancels a proxy thread that is in progress by instructing the GAE to no longer track the specified thread handle as 'In Progress'. The GAE shall discard any transactions associated with threads that are not being tracked as 'In Progress'.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-180. Cancel Proxy Thread Request Payload

Byte Offset	Length in Bytes	Description
0h	2	- Bits[11:0]: PBR-ID: Target PID for the management command - Bits[14:12]: Command Sequence Number: Proxy thread identifier returned by Proxy GFD Management Command request - Bit[15]: Reserved

Table 7-181. Cancel Proxy Thread Response Payload

Byte Offset	Length in Bytes	Description
0h	2	Number of Available Threads : Remaining number of simultaneous proxy operations supported by the GAE.

7.7.15 Global Memory Access Endpoint Management Command Set

This command set is used by the FM to discover and manage the structures and devices involved in providing access to G-FAM and GIM resources.

7.7.15.1 Identify VCS GAE (Opcode 5900h)

This command is used by the FM to query a GAE's capabilities, including maximum number of supported enabled PIDs and maximum number of simultaneous outstanding proxy operations and VendPrefixL0 support. It also reports the remaining number of proxy threads currently available.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-182. Identify VCS GAE Request Payload

Byte Offset	Length in Bytes	Description
0h	1	VCS ID: ID of VCS to query.
1h	1	Start vPPB Instance: Index of vPPB whose FAST Segment Info should be provided in the first entry in vPPB Global Memory Support Info List . The value of 0 represents the GAE. The values of 1 and above represent the DSP vPPBs in increasing Device Number, Function Number order, as defined in Section 7.1.4 .
2h	1	Number of vPPBs: Number of vPPBs whose FAST Segment Info should be provided in vPPB Global Memory Support Info List .

The Identify VCS GAE Response Payload is defined in [Table 7-157](#).

7.7.15.2 Get VCS PID Access Vectors (Opcode 5901h)

This command is used by the FM to query a GAE's current GFD Mapping Vector and VendPrefixL0 Target Vector.

This command will return Invalid Input under the following conditions:

- The requested byte range exceeds the size of the access vector buffer, as defined in [Table 7-164](#)
- The specified VCS does not include a GAE

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-183. Get VCS PID Access Vectors Request Payload

Byte Offset	Length in Bytes	Description
0h	1	VCS ID: ID of VCS to query.
1h	3	Reserved
4h	4	Start Byte: Offset in bytes into Vector Data .
8h	4	Number of Bytes: Size in bytes of Vector Data requested.

The Get VCS PID Access Vectors Response Payload is defined in Table 7-163.

7.7.15.3 Configure VCS PID Access (Opcode 5902h)

This command is used by the FM to control access to a specified PID as reported in the GFD Mapping Vector or VendPrefixLO Target Vector. It is used by the FM to enable or disable access to a PID from a GAE.

Possible Command return codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-184. Configure VCS PID Access Request Payload

Byte Offset	Length in Bytes	Description
0h	1	VCS ID: ID of VCS to configure.
1h	1	Reserved
2h	2	• Bits[11:0]: PID: PID of GFD or VendPrefixLO target • Bits[14:12]: Operation: Specifies which configuration to perform: — 000b = Enable PID access in the GMV — 001b = Disable PID access in the GMV — 010b = Enable PID access in the VTV — 011b = Disable PID access in the VTV — 100b = Update Latency/BW — All other encodings are reserved • Bit[15]: Reserved
4h	8	Latency Entry Base Unit: Latency Entry Base Unit for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Reserved when Operation is 001b or 011b.
Ch	2	Latency Entry: Latency Entry for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Reserved when Operation is 001b or 011b.
Eh	8	BW Entry Base Unit: Bandwidth Entry Base Unit for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Reserved when Operation is 001b or 011b.
16h	2	BW Entry: Bandwidth Entry for path between host and target device, as defined in ACPI HMAT System Locality Latency and Bandwidth Information Structure. Reserved when Operation is 001b or 011b.

7.7.15.4 Get VendPrefixL0 State (Opcode 5903h)

This command is used by the FM to query the enable state of VendPrefixL0 in a VCS. Support for this command indicates whether a PBR switch supports VendPrefixL0. The **Get VendPrefixL0 State** command shall only be implemented by PBR switches that support VendPrefixL0.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-185. Get VendPrefixL0 State Request Payload

Byte Offset	Length in Bytes	Description
0h	1	VCS ID: ID of the VCS to which the GAE or vPPB belongs.

Table 7-186. Get VendPrefixL0 State Response Payload

Byte Offset	Length in Bytes	Description
0h	1	• Bit[0]: Egress Request/Ingress Completion VendPrefixL0 Enabled: Indicates whether support for VendPrefixL0 is enabled (1) or disabled (0) for Egress UIO Requests and Ingress UIO completions in the specified VCS • Bit[1]: Ingress Request VendPrefixL0 Enabled: Indicates whether support for VendPrefixL0 is enabled (1) or disabled (0) for Ingress UIO requests in the specified VCS • Bit[7:2]: Reserved

7.7.15.5 Set VendPrefixL0 State (Opcode 5904h)

This command is used by the FM to enable or disable support for VendPrefixL0 in a VCS. Support for this command indicates whether a PBR switch supports VendPrefixL0; it shall be implemented by and shall only be implemented by PBR switches that support VendPrefixL0.

Possible Command Return Codes:

- Success
- Unsupported
- Invalid Input
- Internal Error
- Retry Required

Command Effects:

- None

Table 7-187. Set VendPrefixL0 State Request Payload

Byte Offset	Length in Bytes	Description
0h	1	VCS ID: ID of the VCS to configure.
1h	1	Reserved
2h	1	- Bit[0]: Enable Egress Request/Ingress Completion VendPrefixL0: Enables (1) or disables (0) support for VendPrefixL0 for Egress UIO Requests and Ingress UIO completions in the specified VCS - Bit[1]: Enable Ingress Request VendPrefixL0: Enables (1) or disables (0) support for VendPrefixL0 for Ingress UIO requests in the specified VCS - Bit[7:2]: Reserved

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